2024 AMC 10B Problem 17 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 17. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 17

Problem:
In a race among 5 snails, there is at most one tie, but that tie can involve any number of snails. For example, the result of the race might be that Dazzler is first; Abby, Cyrus, and Elroy are tied for second; and Bruna is fifth. How many different results of the race are possible?

Answer Choices:

A. 180

B. 361

C. 420

D. 431

E. 720

Solution:

If there are no ties, then there are 5! possible race results. Suppose k of the 5 snails are tied, where $2 \leq k \leq 5$. There are $\binom{5}{k}$ ways to choose the snails that are tied, and then considering those snails as a group, there are $6-k$ entrants and therefore $(6-k)$! orders of finish. The number of possible results is thus

$$5!+\binom{5}{2} \cdot 4!+\binom{5}{3} \cdot 3!+\binom{5}{4} \cdot 2!+\binom{5}{5} \cdot 1!=120+240+60+10+1=(\mathrm{D}) \boxed{431}$$